General References/Resources

(Most resources can be found online and are free)

1. **Course:** Udacity - Intro to Inferential Statistics
2. **Course:** Udacity - Intro to Descriptive Statistics
3. **Course:** Udacity - A/B Testing
4. **Textbook:** Discovering Statistics by Field, Miles
5. **Textbook:** Introduction to Statistical Learning by Gareth James et al.
6. **Presentation:** PyData '15 - Beyond t tests - How to Conclude an Online Experiment Using Python (youtube, slides)
7. **Course Slides:** Stanford's Introduction to Probability and Statistics for Epidemiology (Prof Kristin Sainani) - Introduction to Sample Size and Power Calculations (slides)
8. **Course (Paid):** Stanford's Medical Statistics III (Prof Kristin Sainani) - Common Statistical Tests in Medical Research [many flow chart insights from this course!]
9. **Website:** EvanMiller.org
10. **Website:** www.StatisticsHowTo.com/
11. **Website/Article**: UCLA Stats - What statistical analysis should I use?

LaTex Formula: Z test - Continuous SS

```
n_{1} = \frac{r+1}{r}\:\frac{\sigma^{2}\, (Z_{\beta} + Z_{\alpha/2} )^{2}} {(\overline{x}_{1} - \overline{x}_{2})^{2}}\\ \\

\\\textrm{where}:\\
n_{1} = \textrm{size of smaller group}\\
r = \textrm{ratio of larger group to smaller group}\\
\sigma  = \textrm{standard deviation}\\
\overline{x}_{1} - \overline{x}_{2}= \textrm{difference in means of the outcome}\\
Z_{\beta}   = \textrm{corresponds to power (.84 = 80% power)}\\
Z_{\alpha/2} = \textrm{corresponds two-tailed significance level (1.96 for } \alpha = .05)
```

LaTex Formula: Z test - Binary SS

```
n_{1} = \frac{r+1}{r}\: \frac{\overline{p}(1-\overline{p})\, (Z_{\beta} + Z_{\alpha/2}
)^{2}}{(p_{1} - p_{2})^{2}}\\ \\ \\

\\\textrm{where}:\\
n_{1} = \textrm{size of smaller group}\\
r = \textrm{ratio of larger group to smaller group}\\
\overline{p}(1-\overline{p})  = \textrm{binomial distribution}\\
(p_{1} - p_{2}) = \textrm{difference in means of the outcome}\\
Z_{\beta}   = \textrm{corresponds to power (.84 = 80% power)}\\
Z_{\alpha/2} = \textrm{corresponds two-tailed significance level (1.96 for } \alpha = .05)
```

Please file suggestions/issues here: https://github.com/pleonova/stat-tests or reach out to Paula at p.leonova@gmail.com